The Faculty of Medicine of Harvard University
Curriculum Vitae

Date Prepared:	**April 04, 2024**
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Place of Birth:	Mumbai, India

Education:

2003-2007	B. Pharm	Pharmaceutical Sciences	University of Mumbai, India
2007-2008	Certificate	Course in Intellectual Property	NMIMS University, Mumbai, India.
2008-2014	PhD	Pharmaceutical Sciences (Thesis advisor: Umesh R. Desai)	Virginia Commonwealth University, Virginia, USA.

Postdoctoral Training:

2014-2016	Postdoctoral Scholar	Medicinal Chemistry (Umesh R. Desai)	Virginia Commonwealth University, Virginia, USA.
2017-2021	Research Scholar	Glycomics (Richard D. Cummings)	Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA, USA.

Appointments at Hospitals/Affiliated Institutions:

2017-2021	Research Scholar	Department of Surgery	Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA, USA.
2021-present	Instructor of Surgery, Staff Scientist I	Department of Surgery	Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA, USA.

Committee Service:

Local

2018	Member of the organizing committee for the Harvard Medical School Center for Glycoscience Inaugural Symposium	Harvard Medical School
2019-2021	Communications Chair	BIDMC Postdoctoral Association
2021-2022	Co-director	BIDMC Postdoctoral Association

National

2016	Leader at the 2016 Future GlycoGen Session	NIH Programs of Excellence in Glycosciences Meeting, Bethesda, MD, USA.

International

2019	Member of the organizing committee for the 13th Jenner Glycobiology and Medicine Symposium	13th Jenner Glycobiology and Medicine Symposium, Boston, MA, USA.
2020	Member of the organizing committee for the 12th International Symposium on Glycosyltransferases (GlycoT 2020)	12th International Symposium on Glycosyltransferases (GlycoT 2020), Boston, MA, USA.
2021-present	Member of MIRAGE Commission for Glycan Microarray Analysis	MIRAGE Commission Beilstein Institut

Professional Societies:

2009-present	Phi Kappa Phi honor society	Member
2013-present	Alpha Epsilon Lambda (AEL) honor society	Member
2014-present	Rho Chi honor society	Member
2020-2021	Society for Glycobiology	Member

Editorial Activities:

- **Ad hoc Reviewer**

Chemical Communications
RSC Advances
Journal of Chromatographic Science
Molecules
Analyst
Future Medicinal Chemistry
New Journal of Chemistry
Analytical Methods
Cancers
Journal of Materials Chemistry B
Marine Drugs
Molecular Systems Design & Engineering
Briefings in Bioinformatics
Dalton Transactions
International Journal of Molecular Sciences
Journal of Pharmaceutical and Biomedical Analysis
Nucleic Acids Research
Organic & Biomolecular Chemistry
Scientific Reports
Viruses

- **Other Editorial Roles**

2007-Present	Founding editor	PharmaXChange.info
2023-Present	Associate Editor	Frontiers in Molecular Biosciences - Glycoscience

Honors and Prizes:

2014	2013-2014 Thesis/Dissertation Assistantship	Virginia Commonwealth University	
2014	Best Poster Award	3rd Annual Inter PEG (Programs of Excellence in Glycosciences) National Meeting (Bethesda, MD).	
2014	J. Doyle Smith Award	Department of Medicinal Chemistry (Virginia Commonwealth University)	Outstanding Graduate Studies
2014	Runner-Up Rector and Rorrer Jr. Dean's Award	Department of Medicinal Chemistry (Virginia Commonwealth University)	Excellence in Graduate Studies.
2018	Best Poster Award	256th American Chemical Society National Meeting CARB Poster Session	

Report of Local Teaching and Training

Teaching of Students in Courses:

2010-2012	PHAR 524 Foundations II – Glucose Oxidase Lab	Teaching Assistant
Spring 2010	MEDC 553 Clinical Therapeutics Module I: Introduction to Medicinal Chemistry – Grading	Teaching Assistant
2010-2011	MEDC 533 Pharmacognosy – Grading	Teaching Assistant
Fall 2010	PHAR 523 Foundations I Lab-Drug solubility	Teaching Assistant
2010-2012	PHAR 523 Foundations I Lab-Pharmacognosy	Teaching Assistant
2010-2011	PHAR 640 Foundations V – Grading and Blackboard Maintenance	Teaching Assistant
Fall 2010	PHAR 534 Foundations III– Grading and Blackboard Maintenance	Teaching Assistant
2010-2011	PHAR 556 Clinical Therapeutics Module VI: Neurology I – Grading	Teaching Assistant
2011-2012	MEDC 543 Clinical Chemistry for the Pharmacist – Grading Manuscripts	Teaching Assistant
Spring 2011	PHAR 602 Clinical Therapeutics Module VIII: Psychiatry – Grading	Teaching Assistant
Spring 2011	PHAR 603 Clinical Therapeutics Module IX: Respiratory/Immunology – Grading	Teaching Assistant
Spring 2011	PHAR 645 Foundations VI Lab and Grading	Teaching Assistant
Spring 2016	MEDC 532 Medicinal Chemistry for Nurse Anesthetists – Tutoring	Teaching Assistant

Formal Teaching of Residents, Clinical Fellows and Research Fellows (post-docs):

2013	Workshop: UPLC – MS study of chondroitin di- and oligosaccharides	Programs of Excellence in Glycoscience HSPG-PEG hands-on training session at Virginia Commonwealth University (4 hours)

2015	Workshop: Ultrapressure Liquid Chromatography with Mass Spectrometry.	Virginia Commonwealth University Medicinal Chemistry Graduate Student Association Workshop.

Other Mentored Trainees and Faculty:

2012	Mia Hall – Mentored during their Undergraduate studies as a Summer Intern from Elizabeth City State University
2012-2013	Kim Nguyen – Mentored during their Undergraduate studies under the Virginia Commonwealth University Graduate Student Mentorship Program. Helped identify field of study to pursue.

Report of Regional, National and International Invited Teaching and Presentations

☒ *No presentations below were sponsored by 3rd parties/outside entities*

☐ *Those presentations below sponsored by outside entities are so noted and the sponsor(s) is (are) identified.*

National

2020	Chemical and informatics tools for glycosciences: Glyco-SPOT and GlycoGlyph. Poster Talk. Society for Glycobiology (SFG) Virtual Annual Meeting, Online.

International

2022	From Parallel Synthesis of Glycopeptides, to Glycan Arrays and Glycan Informatics. European Glycoscience Community Webinar Series. Online.

Report of Teaching and Education Innovations

2018-Present	Generated educational videos for the National Center for Functional Glycomics (NCFG) YouTube channel, describing tools to generate and analyze glycan microarray data along with several members of the NCFG.

Report of Technological and Other Scientific Innovations

WO 2014075045 A1. Filed: November, 12, 2013. United States Patent issued; Patent Number: US9758459B2	Allosteric Modulators of Factor XIa as Anticoagulant Agents.
WO 2014194015 A1. Filed: May, 29, 2014. United States Patent issued; Patent Number: US9777123B2.	Sulfated Beta-O4 Low Molecular Weight Lignins.
WO 2015148622 A1. Filed: Mar 25, 2015	A Nanosensor for Assessing Thrombin Inhibitors.

Report of Scholarship

Peer-Reviewed Scholarship in print or other media:

Research Investigations

1. Eckmair, B.; Gao, C.; **Mehta, A. Y.**; Dutkiewicz, Z.; Vanbeselaere, J.; Cummings, R. D.; Paschinger, K.; Wilson, I. B. H., Recognition of highly branched N-glycans of the porcine whipworm by the immune system. Mol. Cell. Proteomics 2024, 23 (2), 100711.
2. **Mehta, A. Y. ***; Tilton, C. A. *; Muerner, L.; Gunten, S.; Heimburg-Molinaro, J.; Cummings, R. D., Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process. Glycobiology 2024, 34(2). (* Co-first author)
3. Goth C.K. *; **Mehta A.Y. ***; McQuillan A.M.; Baker K.J.; Hanes M.S.; Park S.S., Stavenhagen K.; Hjortø G.M.; Heimburg-Molinaro J.; Chaikof E.L.; Rosenkilde M.M.; Cummings R.D., Chemokine binding to PSGL-1 is controlled by O-glycosylation and tyrosine sulfation. Cell Chem Biol. 2023, 30 (8), 893-905 e7. (* Co-first author)
4. Yang, C.; Deng, Y.; Wang, Y.; Xia, C.; **Mehta, A. Y.**; Baker, K. J.; Samal, A.; Booneimsri, P.; Lertmaneedang, C.; Hwang, S.; Flynn, J. P.; Cao, M.; Liu, C.; Zhu, A. C.; Cummings, R. D.; Lin, C.; Mohanty, U.; Niu, J., Heparan Sulfate Glycomimetics via Iterative Assembly of "Clickable" Disaccharides. Chemical Science 2023, 14 (13), 3514-3522.
5. Nielsen, M. A.; Koster, D.; **Mehta, A. Y.**; Stengaard-Pedersen, K.; Busson, P.; Junker, P.; Horslev-Petersen, K.; Hetland, M. L.; Ostergaard, M.; Hvid, M.; Leffler, H.; Kragstrup, T. W.; Cummings, R. D.; Deleuran, B., Increased Galectin-9 Levels Correlate with Disease Activity in Patients with DMARD-Naive Rheumatoid Arthritis and Modulate the Secretion of MCP-1 and IL-6 from Synovial Fibroblasts. Cells 2023, 12 (2).
6. Stavenhagen, K.; **Mehta, A. Y.**; Laan, L.; Gao, C.; Heimburg-Molinaro, J.; van Die, I.; Cummings, R. D., N-glycosylation of Mannose receptor (CD206) regulates glycan binding by C-type lectin domains. J. Biol. Chem. 2022, 102591.
7. Nielsen, M. A.; Juul-Madsen, K.; Stegmayr, J.; Gao, C.; **Mehta, A. Y.**; Greisen, S. R.; Kragstrup, T. W.; Hvid, M.; Vorup-Jensen, T.; Cummings, R. D.; Leffler, H.; & Deleuran, B. W., Galectin-3 Decreases 4-1BBL Bioactivity by Crosslinking Soluble and Membrane Expressed 4-1BB. Front. Immunol. 2022 (13) 915890.
8. Gao, C.; Stavenhagen, K.; Eckmair, B.; McKitrick, T. R.; **Mehta, A. Y.**; Matsumoto, Y.; McQuillan, A. M.; Hanes, M. S.; Eris, D.; Baker, K. J.; Jia, N.; Wei, M.; Heimburg-Molinaro, J.; Ernst, B.; Cummings, R. D., Differential recognition of oligomannose isomers by glycan-binding proteins involved in innate and adaptive immunity. Sci. Adv. 2021, 7 (24).
9. Stavenhagen, K.; Laan, L. C.; Gao, C.; **Mehta, A. Y.**; Heimburg-Molinaro, J.; Glickman, J. N.; van Die, I.; Cummings, R. D., Tumor cells express pauci- and oligomannosidic N-glycans in glycoproteins recognized by the mannose receptor (CD206). Cell. Mol. Life Sci. 2021, 78 (14), 5569-5585.
10. Luetscher, R. N. D.; McKitrick, T. R.; Gao, C.; **Mehta, A. Y.**; McQuillan, A. M.; Kardish, R.; Boligan, K. F.; Song, X.; Lu, L.; Heimburg-Molinaro, J.; von Gunten, S.; Alter, G.; Cummings, R. D., Unique repertoire of anti-carbohydrate antibodies in individual human serum. Sci. Rep. 2020, 10 (1), 15436.
11. **Mehta, A. Y.**; Veeraiah, R. K. H.; Dutta, S.; Goth, C. K.; Hanes, M. S.; Gao, C.; Stavenhagen, K.; Kardish, R.; Matsumoto, Y.; Heimburg-Molinaro, J.; Boyce, M.; Pohl, N. L. B.; Cummings, R. D., Parallel Glyco-SPOT Synthesis of Glycopeptide Libraries. Cell Chem. Biol. 2020, 27 (9), 1207-1219 e9.
12. **Mehta, A. Y.***; Cummings, R. D., GlycoGlyph: A glycan visualizing, drawing and naming application. Bioinformatics 2020, 36 (11), 3613-3614. (*Corresponding Author)
13. Cao, Y.; Park, S. J.; **Mehta, A. Y.**; Cummings, R. D.; Im, W., GlyMDB: Glycan Microarray Database and analysis toolset. Bioinformatics 2020, 36 (8), 2438-2442.
14. York, W. S.; Mazumder, R.; Ranzinger, R.; Edwards, N.; Kahsay, R.; Aoki-Kinoshita, K. F.; Campbell, M. P.; Cummings, R. D.; Feizi, T.; Martin, M.; Natale, D. A.; Packer, N. H.; Woods, R. J.; Agarwal, G.; Arpinar, S.; Bhat, S.; Blake, J.; Castro, L. J. G.; Fochtman, B.; Gildersleeve, J.; Goldman, R.; Holmes, X.; Jain, V.; Kulkarni, S.; Mahadik, R.; **Mehta, A.**; Mousavi, R.; Nakarakommula, S.; Navelkar, R.; Pattabiraman, N.; Pierce, M. J.; Ross, K.; Vasudev, P.; Vora, J.; Williamson, T.; Zhang, W., GlyGen: Computational and Informatics Resources for Glycoscience. Glycobiology 2020, 30 (2), 72-73.
15. Wei, M.; McKitrick, T. R.; **Mehta, A. Y.**; Gao, C.; Jia, N.; McQuillan, A. M.; Heimburg-Molinaro, J.; Sun, L.; Cummings, R. D., Novel Reversible Fluorescent Glycan Linker for Functional Glycomics. Bioconjug. Chem. 2019, 30 (11), 2897-2908.

16. **Mehta, A. Y.***; Cummings, R. D., GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays. Bioinformatics 2019, 35 (18), 3536-3537. (*Corresponding Author)
17. Byrd-Leotis, L.; Gao, C.; Jia, N.; **Mehta, A. Y.**; Trost, J.; Cummings, S. F.; Heimburg-Molinaro, J.; Cummings, R. D.; Steinhauer, D. A., Antigenic Pressure on H3N2 Influenza Virus Drift Strains Imposes Constraints on Binding to Sialylated Receptors but Not Phosphorylated Glycans. J. Virol. 2019, 93 (22).
18. Mangrum, J. B.; **Mehta, A. Y.**; Alabbas, A. B.; Desai, U. R.; Hawkridge, A. M., Comparative analysis of INLIGHT-labeled enzymatically depolymerized heparin by reverse-phase chromatography and high-performance mass spectrometry. Anal. Bioanal. Chem. 2017, 409 (2), 499-509.
19. Verespy, S., 3rd; **Mehta, A. Y.**; Afosah, D.; Al-Horani, R. A.; Desai, U. R., Allosteric Partial Inhibition of Monomeric Proteases. Sulfated Coumarins Induce Regulation, not just Inhibition, of Thrombin. Sci. Rep. 2016, 6, 24043.
20. **Mehta, A. Y.**; Mohammed, B. M.; Martin, E. J.; Brophy, D. F.; Gailani, D.; Desai, U. R., Allosterism-based simultaneous, dual anticoagulant and antiplatelet action: allosteric inhibitor targeting the glycoprotein Ibalpha-binding and heparin-binding site of thrombin. J. Thromb. Haemost. 2016, 14 (4), 828-38.
    - Highlighted by the editorial section in the journal DOI: 10.1111/jth.13308
21. Xu, G. G.; Deshpande, T. M.; Ghatge, M. S.; **Mehta, A. Y.**; Omar, A. S.; Ahmed, M. H.; Venitz, J.; Abdulmalik, O.; Zhang, Y.; Safo, M. K., Design, Synthesis, and Investigation of Novel Nitric Oxide (NO)-Releasing Prodrugs as Drug Candidates for the Treatment of Ischemic Disorders: Insights into NO-Releasing Prodrug Biotransformation and Hemoglobin-NO Biochemistry. Biochemistry 2015, 54 (49), 7178-92.
22. Boothello, R. S.; Sarkar, A.; Tran, V. M.; Nguyen, T. K.; Sankaranarayanan, N. V.; **Mehta, A. Y.**; Alabbas, A.; Brown, S.; Rossi, A.; Joice, A. C.; Mencio, C. P.; Quintero, M. V.; Kuberan, B.; Desai, U. R., Chemoenzymatically prepared heparan sulfate containing rare 2-O-sulfonated glucuronic acid residues. ACS Chem. Biol. 2015, 10 (6), 1485-94.
23. **Mehta, A. Y.**; Thakkar, J. N.; Mohammed, B. M.; Martin, E. J.; Brophy, D. F.; Kishimoto, T.; Desai, U. R., Targeting the GPIbalpha binding site of thrombin to simultaneously induce dual anticoagulant and antiplatelet effects. J. Med. Chem. 2014, 57 (7), 3030-9.
    - Highlighted in SciBX 7(14); doi:10.1038/scibx.2014.401
24. **Mehta, A. Y.**; Desai, U. R., Substantial non-electrostatic forces are needed to induce allosteric disruption of thrombin's active site through exosite 2. Biochem. Biophys. Res. Commun. 2014, 452 (3), 813-6.
25. Argade, M. D.*; **Mehta, A. Y.***; Sarkar, A.; Desai, U. R., Allosteric inhibition of human factor XIa: discovery of monosulfated benzofurans as a class of promising inhibitors. J. Med. Chem. 2014, 57 (8), 3559-69. (* Co-first author)
26. Sidhu, P. S.; Abdel Aziz, M. H.; Sarkar, A.; **Mehta, A. Y.**; Zhou, Q.; Desai, U. R., Designing allosteric regulators of thrombin. Exosite 2 features multiple subsites that can be targeted by sulfated small molecules for inducing inhibition. J. Med. Chem. 2013, 56 (12), 5059-70.
27. Al-Horani, R. A.; Ponnusamy, P.; **Mehta, A. Y.**; Gailani, D.; Desai, U. R., Sulfated pentagalloylglucoside is a potent, allosteric, and selective inhibitor of factor XIa. J. Med. Chem. 2013, 56 (3), 867-78.
28. Al-Horani, R. A.; **Mehta, A. Y.**; Desai, U. R., Potent direct inhibitors of factor Xa based on the tetrahydroisoquinoline scaffold. Eur. J. Med. Chem. 2012, 54, 771-83.
29. Sidhu, P. S.; Liang, A.; **Mehta, A. Y.**; Abdel Aziz, M. H.; Zhou, Q.; Desai, U. R., Rational design of potent, small, synthetic allosteric inhibitors of thrombin. J. Med. Chem. 2011, 54 (15), 5522-31.
30. Desai, B. J.; Boothello, R. S.; **Mehta, A. Y.**; Scarsdale, J. N.; Wright, H. T.; Desai, U. R., Interaction of thrombin with sucrose octasulfate. Biochemistry 2011, 50 (32), 6973-82.

Other peer-reviewed scholarship

1. **Mehta, A. Y.**; Heimburg-Molinaro, J.; Cummings, R. D., Tools for generating and analyzing glycan microarray data. Beilstein J. Org. Chem. 2020, 16, 2260-2271.
2. **Mehta, A. Y.**; Heimburg-Molinaro, J.; Cummings, R. D.; Goth, C. K., Emerging patterns of tyrosine sulfation and O-glycosylation cross-talk and co-localization. Curr. Opin. Struct. Biol. 2020, 62, 102-111.
3. **Mehta, A. Y.**; Jin, Y.; Desai, U. R., An update on recent patents on thrombin inhibitors (2010 - 2013). Expert Opin. Ther. Pat. 2014, 24 (1), 47-67.

Reviews, chapters, monographs and editorials

1. Desai, U. R.; Al-Horani, R. A.; **Mehta, A. Y.**; Abdel Aziz, M. H. Cardiovascular Agents. In *The Chemistry and Physics of Drugs Used in Anesthesia*. 2nd Ed. Kier, L. B.; Dowd, C. S.; Eds. AANA Publishing: Park Ridge, IL 2014. (ISBN: 9780982991220).

Thesis:

Mehta, A.Y. 2014, Synthetic, sulfated, lignin-based anticoagulants, Virginia Commonwealth University.

Abstracts, Poster Presentations and Exhibits Presented at Professional Meetings:

1. From Parallel Synthesis of Glycopeptides, to Glycan Arrays and Glycan Informatics. **Mehta A. Y.** (2022) European Glycoscience Community Webinar Series, Online.
2. Chemical and informatics tools for glycosciences: Glyco-SPOT and GlycoGlyph. **Mehta A. Y.**, Veeraiah R. H., Goth C. K., Hanes M. S., Gao C., Matsumoto Y., Heimburg-Molinaro J. H., Pohl N. L., Cummings R. D. (2020) Society for Glycobiology (SFG) Virtual Annual Meeting, Online. (Selected for Poster Talk)
3. Use of glycopeptide libraries to deduce specificity glycosyl transferases ST6GalNAc1, T-Synthase and GCNT1. **Mehta A. Y.**, Goth C. K., Hanes S. H., Veeraiah R. H., Gao C., Stavenhagen K., Pohl N. L., Cummings R. D. (2020) 12th International Symposium on Glycosyltransferases, Online.
4. GLAD (GLycan Array Dashboard) - Visualizing Glycan Microarray Data in an Interactive Manner. **Mehta A. Y.**, Cummings R. D. (2019) 13th Jenner Glycobiology and Medicine Symposium, Boston, MA.
5. Microgram scale synthesis of Glycopeptide Libraries: Use as standards and in screening. **Mehta A. Y.**, Dutta S., Hanes M. S., Veeraiah R. H., Gao C., Matsumoto Y., Pohl N. L., Boyce M., Cummings R. D., (2018) 256th American Chemical Society National Meeting & Exposition, Boston, MA.
    - CARB Poster Session. (Best Poster Award)
6. First Steps Towards the Development of a Gold Nanosensor for Detection and Quantification of Antithrombin. **Mehta A. Y.**, Barlow G. N., Desai U. R. (2014) School of Pharmacy Research and Career Day Meet, Richmond, VA.
7. Targeting Overlapping Glycosaminoglycan and Sulfated Tyrosine Binding Sites on Thrombin to Induce Allosteric Antithrombotic and Competitive Antiplatelet Activities. **Mehta A. Y.**, Thakkar J. N., Mohammed B. M., Martin E. J., Brophy D. F., Kishimoto T., Gailani D., and Desai U. R. (2014) 3rd Annual Inter-PEG (Programs of Excellence in Glycosciences) Meeting, Bethesda, MD.
    - Best Poster Award.
8. Targeting the GPIbα binding site of thrombin for dual anticoagulant and antiplatelet activity. **Mehta A. Y.**, Thakkar J. N., Mohammed B. M., Martin E. J., Brophy D. F., Kishimoto T., Gailani D. and Desai U. R. (2013) School of Pharmacy Research and Career Day Meet, Richmond, VA.
9. Prophylactic possibilities of sulfated lignin as anticoagulant via heparin-like allosterism against thrombin. **Mehta A. Y.**, Thakkar J. N., Martin E. J., Brophy D. F., Kishimoto T., Gailani D., and Desai U. R. (2013) 2nd Annual Inter-PEG (Programs of Excellence in Glycosciences) Meeting, Cleveland, OH.
10. Chemically synthesized sulfated-β-O4-lignin (SbO4L) polymer with selective allosteric inhibition of thrombin. **Mehta A. Y.**, Thakkar J. N., Martin E. J., Brophy D. F., Kishimoto T. and Desai U. R. (2013) Drug Discovery and Therapy World Congress, Boston, MA.
11. Sulfated beta-O-4 lignin (SbO4L): An anticoagulant with selective allosteric modulation of thrombin. **Mehta A. Y.**, Thakkar J. N., Martin E. J., Brophy D. F., Kishimoto T. and Desai U. R. (2012) 29th Annual Daniel E. Watts Research Poster Symposium, Richmond, VA.
12. Studies on sulfated β-O-4 lignin (SbO4L) as a selective, allosteric modulator of thrombin. **Mehta A. Y.**, Thakkar J. N., Martin E. J., Brophy D. F., Kishimoto T. and Desai U. R. (2012) School of Pharmacy Research and Career Day Meet, Richmond, VA.
    - Runner-Up Award
13. Development of reverse-phase ion-pairing ultraperformance liquid chromatography/mass spectrometry (RPIP-UPLC/MS) methods for fingerprinting sulfated small molecules. Ponnusamy P., **Mehta A. Y.**, Karuturi R., Sidhu P. S., Al-Horani R. A. and Desai U. R. (2012) 15th Annual Graduate Research Symposium, Richmond, VA.
14. High-throughput screening and discovery of allosteric factor XIa inhibitors. **Mehta A. Y.**, Argade M. D. and Desai U. R. (2011) 63rd SERMACS (Southeastern Regional Meeting of the American Chemical Society), Richmond, VA.

15. Sulfated β-O4 low molecular weight lignins are potent and specific inhibitors of thrombin. **Mehta A. Y.**, Thakkar J. N. and Desai U. R. (2010) 12th Annual Graduate Student Symposium, VCU, Richmond, VA.

Narrative Report

I am a scientist who is trained in pharmaceutical sciences with a major in medicinal chemistry, having a strong interest in glycosciences, high throughput screening platforms (e.g., microarrays, glycopeptide libraries, Luminex platform) and bioinformatics. Glycans are polymers of sugars and are an important class of polymers found in all living cells. They are known to impart a number of critical biological functions. Through my broad expertise, I have managed to create new microarray platforms which present glycans on various probes, to interrogate various biological interactions. Microarray platforms allow the rapid and efficient screening of glycan-protein interactions which could be useful in a number of conditions, including infectious diseases, cancer, inflammation and immunology.

Additionally, my background in medicinal chemistry enabled me to develop new synthetic methodology to produce glycopeptide libraries in a rapid and parallel manner. This led to development of more diverse and biologically-relevant microarrays which provide useful tools for the field of glycoscience.

I have also produced bioinformatics software to draw representations of glycans in symbol nomenclature used commonly in the field, and provide a way to automatically produce nomenclature for such structures. I have also produced web-based software to help with the data analysis and data visualization of glycan microarrays. I am currently developing new bioinformatics tools for data management and capturing various parameters of glycan microarray and glycomics experiments, in order to make the data more searchable and accessible to advanced data analysis e.g., machine learning.

I am passionate about being a scientist and developing new tools to automate and miniaturize science and find new ways to help others.